

Catalog Search

Entire Catalog

Search Catalog


☐ Whole Word/Phrase

Advanced Search

Catalog Home

Courses

Academic Programs

Schools & Departments

Dialogues Curriculum

Dialogues Requirements

Outline of Dialogues Requirement

HB 1606: Graduate Outcomes

General Information

The Campus

Admission

Fees and Financial Aid

Academic Policies and Procedures

Truman Policies

Transferring In and Testing Out

Student Services

Student Involvement

Special Academic Programs

Honors

Graduation

Graduate Studies

Board of Governors and Administrators

Emeriti

Faculty

Staff

Former Catalogs

My Portfolio

Biochemistry and Molecular Biology (BS)


[Return to: Catalog Search](#)

Affiliation: School of Science and Mathematics

THE MAJOR

The Biochemistry and Molecular Biology degree program aims to provide students with the knowledge necessary for a comprehensive understanding of how chemical processes and molecular mechanisms impact biological systems. To that end, the curriculum described here includes a variety of courses across chemistry and biology disciplines: cell biology and genetics act as a foundation in biochemistry, organic and biophysical chemistry provide an understanding of relevant reactions and molecular interactions, and upper-level electives in biochemistry and molecular biology provide a focused, in-depth view of specialized topics. Through this carefully designed curriculum, students will develop the critical thinking, analytical, and technical skills necessary to be creative, effective, and responsible scientists. Biochemistry and molecular biology represents a growing interdisciplinary field, and when coupled with a strong liberal arts education, students graduating with this degree would be well-prepared for a wide range of careers in industry or professional and graduate schools.

BIOCHEMISTRY AND MOLECULAR BIOLOGY COMMUNICATION

All biochemistry and molecular biology majors will develop their skills as speakers and as writers. Beginning in introductory courses, students learn the fundamentals of recordkeeping and the dissemination of laboratory data through the completion of detailed laboratory notebooks. **CHEM 345 - Junior Chemistry Seminar** focuses on the mechanisms chemists use to disseminate information, both orally and in print, with students preparing and delivering an oral presentation. **CHEM 312 - Foundations of Chemical Analysis** and **CHEM 326 - Quantum Mechanics and Spectroscopy Laboratory** are writing-enhanced courses where students prepare multiple formal laboratory reports, some in the format of an article in a professional journal. Many other BIOL and CHEM courses involve formal and informal writing and speaking experiences in a variety of settings.

DEPARTMENTAL HONORS IN BIOCHEMISTRY AND MOLECULAR BIOLOGY:

Students meeting each of the Academic and Research requirements below will be awarded Departmental Honors.

Academic Requirements for Departmental Honors

1. Upon graduation, the student must have a cumulative GPA of 3.50 or greater.
2. The student must score at or above the 75th percentile on the American Chemical Society nationally-normed exam for a two-semester sequence of Biochemistry.

Research Requirements for Departmental Honors

1. The student must design and conduct an original research project in consultation with a Truman Biology or Chemistry faculty member. If the research is conducted outside of Truman (e.g., at A.T. Still University (ATSU) or a summer program), a consulting Truman Biology or Chemistry faculty member must be involved from the outset.
2. The student must complete a written manuscript of this research that is of publishable quality and in journal style. The manuscript must be approved by a committee of at least three Truman biology and Chemistry faculty.
3. The student must present a 20 minute seminar on the research at a regular Biology or Chemistry weekly seminar. See your academic advisor or the Department Chairs for specific details.

ACS CERTIFICATION

Students completing the Bachelor of Science program in Biochemistry and Molecular Biology as described in this catalog meet the American Chemical Society (ACS) requirements for certification upon graduation (pending approval). Specific questions about the ACS certification requirements may be addressed to the Chair of the Department of Physical Sciences.

DEGREE REQUIREMENTS:

Dialogues Requirements: 42-61 Credits

Missouri Statute: 0-4 credits

Bachelor of Science Requirement: 8-10 Credits

- **PHYS 185 - College Physics I** Credit(s): 4
OR **PHYS 195 - Physics I** Credits: 5
- **PHYS 186 - College Physics II** Credit(s): 4
OR **PHYS 196 - Physics II** Credits: 5

MAJOR REQUIREMENTS:

The Biochemistry and Molecular Biology major consists of two (2) parts: Required Support and Major Requirements. Each student must complete both parts.

Part I: Required Support: 13-16 Credits

- **CHEM 130 - Chemical Principles I** Credit(s): 4

- **MATH 198 - Analytic Geometry and Calculus I** Credit(s): 5
OR (MATH 187 - Calculus I with Precalculus Review I Credit(s): 4 **and MATH 197 - Calculus I with Precalculus Review II** Credit(s): 4)
- **MATH 263 - Analytic Geometry and Calculus II** Credit(s): 4

Part II: Major Requirements: 55 Credits

- **BCMB 145 - Freshman Biochemistry and Molecular Biology Seminar** Credit(s): 1
OR CHEM 145 - Freshman Chemistry Seminar Credit(s): 1
- **CHEM 131 - Chemical Principles II** Credit(s): 4
- **BIOL 107 - Cells, Molecules, and Genes** Credit(s): 4
- **CHEM 245 - Sophomore Chemistry Seminar** Credit(s): 1
- **CHEM 225 - Foundations of Inorganic Chemistry with Lab** Credit(s): 4
- **CHEM 335 - Biochemistry I: Structure and Function** Credit(s): 3
- **BIOL 300 - Genetics** Credit(s): 4
- **BIOL 330 - Cell Biology** Credit(s): 4
- **CHEM 329 - Organic Chemistry I** Credit(s): 3
- (**CHEM 330 - Organic Chemistry I Lab** Credit(s): 1 **and CHEM 332 - Organic Chemistry II Lab** Credit(s): 1)
OR CHEM 333 - Organic Chemistry Lab Credit(s): 2
- **CHEM 331 - Organic Chemistry II** Credit(s): 3
- **CHEM 345 - Junior Chemistry Seminar** Credit(s): 1
- **CHEM 312 - Foundations of Chemical Analysis** Credit(s): 5
- **CHEM 323 - Thermodynamics and Kinetics** Credit(s): 3
- **CHEM 324 - Thermodynamics and Kinetics Laboratory** Credit(s): 2
- **BCMB 445 - Senior Biochemistry and Molecular Biology Capstone Seminar** Credit(s): 1

Electives

At least ten credits of electives are required, including at least one lab credit.

At least five hours must come from the list below:

- **BCMB 443 - Biochemical Research** Credit(s): 3
- **BIOL 305 - Virology** Credit(s): 3
- **BIOL 331 - Introduction to Evolutionary Biology** Credit(s): 4
- **BIOL 333 - Introduction to Neurobiology** Credit(s): 3
- **BIOL 334 - Molecular Pharmacology** Credit(s): 3
- **BIOL 362 - Embryology and Developmental Biology** Credit(s): 4
- **BIOL 370 - Introduction to Bioinformatics** Credit(s): 3
- **BIOL 502 - Biometry** Credit(s): 3
- **BIOL 513 - Microbial Genetics** Credit(s): 4
- **BIOL 530 - Advanced Cell Biology** Credit(s): 4
- **CHEM 310 - Biochemistry Laboratory** Credit(s): 2
- **CHEM 435 - Biochemistry II: Metabolism** Credit(s): 3

- **CHEM 518 - Advanced Topics (Topic)** Credit(s): 1 to 4 (each topic)
No more than 4 credits total of CHEM 518 may be used.
- **STAT 331 - Biostatistics** Credit(s): 3

Remaining elective credit can be fulfilled by any BCMB, BIOL, or CHEM courses at the 300 level or above.

Electives to Total: 120 Credits

Note:

Students majoring in Biochemistry and Molecular Biology (BS) may not also earn a major or minor in Biology or Chemistry.

Further Degree Criteria:

Any student obtaining a bachelor's degree in Biochemistry and Molecular Biology from Truman must:

1. Complete 12 credits of 300-level or above chemistry or biology credits at Truman;
2. Achieve a minimum GPA of 2.0 in all courses listed as Major Requirements, Required Support, and Bachelor of Science requirements, with no more than a total of 5 credits of "D" across those areas;
3. Score at or above the 20th percentile on the BCMB senior exam;
4. Complete the University's Senior Portfolio Project.

← Return to: [Catalog Search](#)

